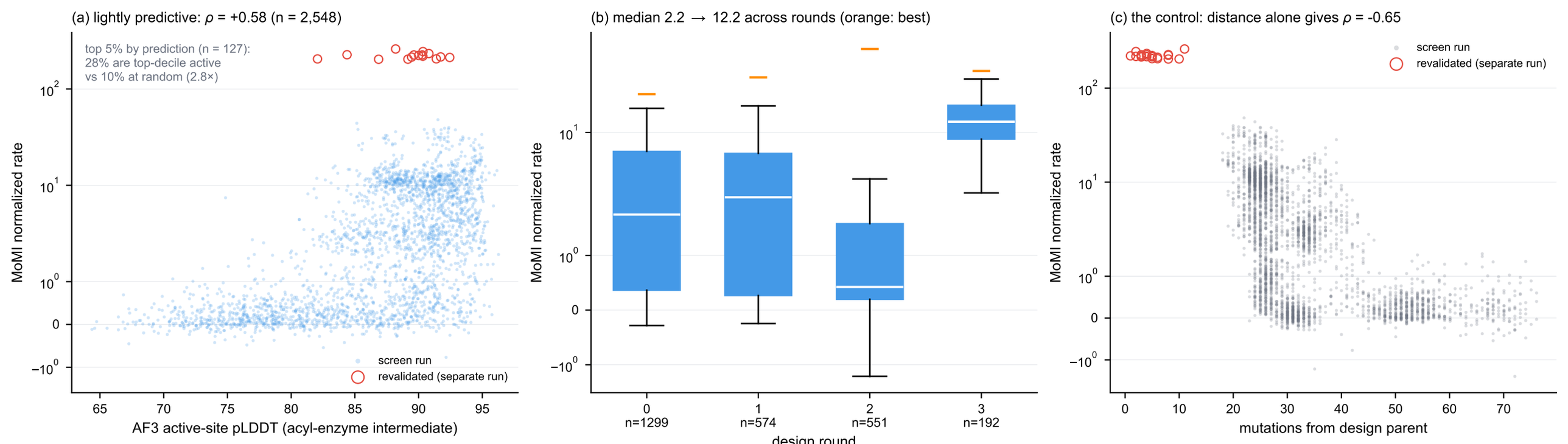

(a) lightly predictive: ρ = +0.58 (n = 2,548)
top 5% by prediction (n = 127):
28% are top-decile active
vs 10% at random (2.8×)
MoMI normalized rate
AF3 active-site pLDDT (acyl-enzyme intermediate)
screen run
revalidated (separate run)
(b) median 2.2 → 12.2 across rounds (orange: best)
MoMI normalized rate
0
n=1299
1
n=574
2
n=551
3
n=192
design round
(c) the control: distance alone gives ρ = -0.65
MoMI normalized rate
mutations from design parent
screen run
revalidated (separate run)